

Agilent Resolve Tactical Analyzer

Raman handheld tactical analyzer for through-barrier identification





Tactical through-barrier identification without compromise

Built for rapid, handheld detection for demanding environments, the Agilent Resolve Raman tactical analyzer offers you the following benefits:

- **Insight:** Confident detection through opaque and colored barriers
- **Safety:** Through-barrier identification without requiring containers to be disturbed or opened
- **Speed:** Accurate identification of chemicals and mixtures in approximately 40 seconds, with no need for sample preparation or consumables
- **Ruggedness:** Robust device built to withstand tough environments
- **Ease of use:** Large buttons and simple interface designed for easy operation in protective clothing

Proven capability for hazardous materials, explosive ordnance disposal, and law enforcement

The Resolve tactical analyzer is the world's only handheld Raman system that enables true through-barrier identification of controlled, hazardous, or contraband materials. The system rapidly detects and identifies materials from comprehensive libraries including explosives,

precursors, toxic industrial chemicals, chemical warfare agents, and narcotics. Unique to Agilent, our proprietary handheld spatially offset Raman spectroscopy (SORS) technology enables positive identification through a wide range of sealed, nonmetallic containers, barriers, and packaging.



Hazardous materials



Search and law enforcement



Military and explosive ordnance disposal



Customs, borders, mailroom, and checkpoint screening

Proven capability in chemical, narcotic, and explosive detection

Resolve differs from conventional handheld Raman identification systems, which typically only operate with line-of-sight measurements of the sample under investigation.

Analyze solids, powders, liquids, and mixtures, such as:



Industrial chemicals



Household chemicals



Spills



Explosives



Narcotics



National security threats

Versatile measurement tools

Resolve can act as a conventional handheld Raman system with point-and-shoot and vial holder measurements—plus, the SORS technology adds the capability to identify materials through barriers.

The system works in three modes:



Through-Barrier mode

Identifies through nonmetallic, sealed containers such as colored and opaque plastics, glass, paper, wrapping, sacks, and fabrics.



Surface Scan mode

Uses line-of-sight measurements, similar to conventional Raman identification systems.



Vial Holder mode

Quickly identifies materials contained within glass vials in a custom holder.



True through-barrier detection and identification

Conventional handheld Raman systems are typically limited to operation through clear plastic bags or glass vials. If materials are concealed behind thicker, colored, or opaque barriers, it might be necessary to open and take a sample. In hazardous material response scenarios, opening or disturbing containers can increase risk to the operator and, sometimes, the public. Taking samples can also unnecessarily disturb crime scene evidence.

Contrastingly, the Resolve system's comprehensive through-barrier capabilities enable response teams to identify container contents early in an operation, before escalation. This allows information about the situation to be gathered quickly and efficiently, enabling better critical decision-making.

Identification through sealed containers provides:

- **Safer operation:** Keeps hazards contained
- **Efficiency:** Removes the need to take samples, move objects, and waste valuable time in personal protective equipment
- **Preservation of evidence:** Keeps crime scenes intact with containers undisturbed
- **Faster decision-making:** Complete, accurate information is obtained earlier in the operation

Resolve enables identification through containers such as:



Colored plastic



Colored glass



Paper



Sacks



Packaging, padded envelopes, and fabric



Increasing the power of handheld Raman identification

Agilent Resolve Raman

Resolve extends through-barrier capability to work with a wide range of nonmetallic containers.



Conventional Raman identification systems

Conventional Raman systems require line-of-sight so work with clear plastic bags and vials, and some translucent packaging.



■ Usually routine

■ Usually more difficult*

* Successful identification depends on container/contents combination.

Intuitive interface, clear results

Resolve combines a tough handheld unit with a simple, user-friendly interface. Ruggedized and hardened, the system is built to withstand harsh environments. The software interface is simple to use; full-system control requires only seven large, responsive buttons, which are ideal for gloved use—even in Level A PPE. Operational managers can fully customize workflows, laser-arming passwords, and the format of results and associated metadata.

Removable nose cone for contact and noncontact scan modes:

In-field calibration and performance check built into a protective nose cap

Large 12 cm (4.7 in) display:

Clear, high-contrast graphics for use in all light conditions

Large buttons with positive feedback:

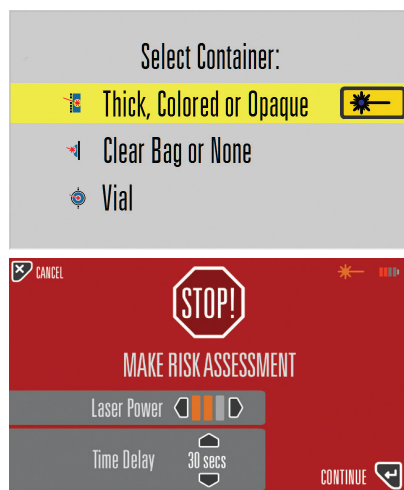
Designed for ease of use with gloves

Li ion battery pack:

Enables up to four hours of continuous use



All accessories, cables, and chargers are supplied within a hard transport case



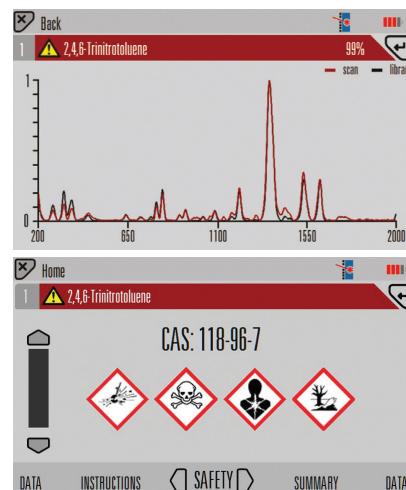
Measurement

Conduct through-barrier, surface, or vial holder scans. Measurements are completed in under 60 seconds. Operators can reduce laser power and set a scan delay where necessary.



Results

Display results as BEST match only, ALL results, or user-defined PRIORITIES only.



Analysis

Interrogate any result to see spectral views and simple, user-friendly chemical data.

Overplot and compare library spectra. Search by chemical name or CAS number.

Intuitive interface, clear results



Removable Wi-Fi access provides the ability to perform remote measurements for increased safety. Reachback requests can be sent directly to Agilent through Wi-Fi.



Users can export single scan data files to USB (or via Wi-Fi) for offline analysis in CSV or SPC formats; export single scan evidentiary report PDF; and export reachback files for specialist support.



Resolve scan data can be used to create new library items on the spot using the onboard user libraries. For more advanced library creation functions, choose Agilent Command Fleet Management software.



An additional automatic self-test run ensures data quality and detection performance.



Overplot previous scans or any library item to obtain further information on your scan.

Resolve system specifications

Specification	Description
Dimensions	Width 155 mm (6.1 in)
	Height 290 mm (11.4 in)
	Depth 73 mm (2.9 in)
Weight	2.2 kg (4.9 lb) including battery
Modes of operation	Through-barrier scan
	Surface scan (conventional point and shoot Raman)
	Vial Holder mode
Additional benefits	830 nm laser—allows scanning of fluorescent materials
	Unique optics—safer operation with sensitive samples
Safety	Scan delay
	Adjustable laser power (475 mW maximum)
Survivability	Shock, drop, and vibration tested to international and military standards (further details available on request)
	IP67—ingress protection against dust and water
Accessories	Calibration piece, laser safety glasses, shoulder strap, and carry case
	Two Li ion battery packs and one single-bay charger
	Vial holder
Power requirements	Rechargeable Li ion battery
	Mains adaptor
Connectivity	USB 2.0
	Wi-Fi (removable)

Command Fleet Management software

The Agilent Command Fleet Management software is designed to enhance the use of your Resolve tactical analyzer, whether you're an operator or a fleet manager. Each module is designed to cover your needs with simple, easy-to-follow workflows. Two modules are currently available: Field User and Management. The modules work seamlessly together with Resolve for library creation, management, and deployment.



Software features



Easily create batch or single files from Resolve databases. File types: CSV, PDF, SPC.



Send batch reachback files to Agilent for assistance with spectra and results.



Change passcodes, settings, and active libraries for any Resolve in your fleet.



Use Resolve scan data to create new library items.



Add hazard and chemical information to be displayed on Resolve's metadata pages. Create custom hazard classes.



Package new library items into libraries and distribute to any Resolve in your fleet.

Services

Reachback

Agilent has a team of scientists on hand to aid your mission.

Resolve customers have access to the Agilent reachback service, which offers advice and confirmation on your sample identification. The reachback team of skilled operators are spectroscopically trained to detect potential threats and flag anything suspicious. Simply email your reachback file to **reachback@agilent.com** or upload to the [scan upload page](#), providing as much information as possible.

More information about the reachback service can be found here: [Agilent Reachback Service for Resolve](#).



Library updates

The [Resolve tactical handheld Raman analyzer](#) is provided with fully loaded, ready-to-use libraries, including chemical and safety metadata. The large onboard library enables rapid identification of a wide range of materials.

With threats continuously developing, [Resolve libraries](#) are updated every six months to keep detection up to date with emerging threats. Each library update focuses on new psychoactive substances, narcotics, excipients, explosives, and benign chemicals, so your instrument can detect and identify the most current threats.



Onsite training

Agilent offers dedicated Resolve training to ensure the best performance can be achieved by any user. Whether you need a standard user session, a “train the trainer” course, refresher training, or a customer-specific symposium, Agilent is here to provide support for your team.

A typical training session includes:

- General Resolve usage training
- Guidance on measurement and ID to allow for faster critical decisions
- Safe operation procedures
- Tips and tricks tailored to your needs
- Hands-on session with user samples. Additional Resolve units provided, if less than one unit per three trainees is available onsite

Contact cls.support@agilent.com for more information



Learn more:

www.agilent.com/chem/raman

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WARNING – INVISIBLE LASER RADIATION
AVOID EXPOSURE TO BEAM
CLASS 3B LASER PRODUCT
(IEC/EN 60825-1 / 2014)
MAX. OUTPUT: < 475 mW / WAVELENGTH 830 nm

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